Dhruv Dighrasker

dhruvdi@umich.edu | linkedin.com/in/dhruv-dighrasker | dighrasker.github.io/dhruv

Education

University of Michigan - College of Engineering **Ann Arbor, MI**

M.S.E in Electrical and Computer Engineering (Integrated Circuits & VLSI) *Apr. 2027*

- **GPA/Coursework:** 4.0/4.0 — Parallel Computer Architecture, VLSI, Quantum Hardware, Monolithic Amplifiers

B.S.E in Computer Engineering *Dec. 2025*

- **GPA/Coursework:** 3.64/4.0 — Advanced Computer Architecture, Operating Systems, GPU Programming, Digital Integrated Circuits, Digital Signal Processing, Data Structures + Algorithms, Digital Logic Design, Electrical Circuits

Technical Skills

Languages: SystemVerilog/Verilog, C, C++, Python, ARM Assembly, MATLAB, Tcl, Bash
Tools: Synopsys VCS, Design Compiler, Cadence Xcelium, Verisium Debug, Verdi, vManager, IMC, Quartus, Virtuoso, Git
Verification: UVM, SystemVerilog Assertions, UVM RAL

Work Experience

Renesas Electronics **Duluth, GA**

Design Verification Intern *May 2026 - Aug. 2026*

- Built directed UVM stimulus using Bash sweeps across 408 back-to-back DDR command configurations
- Implemented SystemVerilog coverpoints and cross coverage for DDR interamble timing from +4 UI to -52 UI across 12 pre/postamble modes, measuring scenario coverage across 3,000-seed regressions in Cadence vManager
- Refined the coverage model through waveform-level analysis in Verisium Debug, classifying 90+ cross combinations as unreachable and closing three valid coverage holes to raise functional coverage from 82% to 98%

Computer Engineering Laboratory **Ann Arbor, MI**

Research Assistant *Feb. 2025 - Present*

- Built a configurable TAGE branch-predictor simulator in C to process 100+ CBP2025 traces and generate reference accuracy, MPKI, and misprediction-rate metrics for Transformer-based input-feature ablations
- Designing feature-aware folding and hashing schemes to integrate the highest-value context-path, local-history, local-bias, and misprediction-feedback inputs into TAGE

University of Michigan - CSE Department **Ann Arbor, MI**

Graduate Student Instructor (former TA), Digital Logic Design *Apr. 2024 – Present*

- Led FPGA-based labs for 220+ student course on FSMs and timing analysis using Verilog, Quartus, and ModelSim
- Maintained Verilog autograder infrastructure for 8+ FPGA projects, validating student RTL and running mutation-test scripts against student-written testbenches

Projects

Superscalar Out-of-Order RISC-V CPU

- Designed a 32-bit, parameterizable N-way superscalar, R10K-style out-of-order RISC-V processor in SystemVerilog
- Integrated register renaming, nonblocking caches, tournament branch prediction, and LSQ with byte-level forwarding
- Verified functionality by developing custom testbenches and achieved performance of 1.13 IPC for clock period of 7.8 ns

Multi-Core L2 Cache with UVM Verification

- Designed parameterized SystemVerilog RTL for a 4-core shared write-back L2 cache with 512 KB capacity, 8-way set associativity, 64-byte cache lines, MESI coherence, 8 MSHRs, and an AXI4-Lite DRAM interface
- Built a multi-agent UVM environment with a coherence-aware scoreboard, validating functionality across 90+ directed regression tests targeting MESI transitions, dirty evictions, refills, write-backs, and MSHR saturation

FlashAttention-2 ASIC Accelerator

- Designed fixed-point FlashAttention-2 ASIC accelerator with tiled, I/O-aware dataflow and fully fused attention pipeline
- Implemented Expmul-based softmax and a Q-format datapath achieving 385 MHz with 1.02-LSB RMSE and 4-LSB maximum error versus a floating-point reference

SystemVerilog RTL IP Library

- Developed reusable RTL IP library with parameterized components for processor datapaths & SoC-style systems
- Implemented RV32 integer ALU, synch/asynch FIFOs, & configurable arbiters with clean hardware interfaces